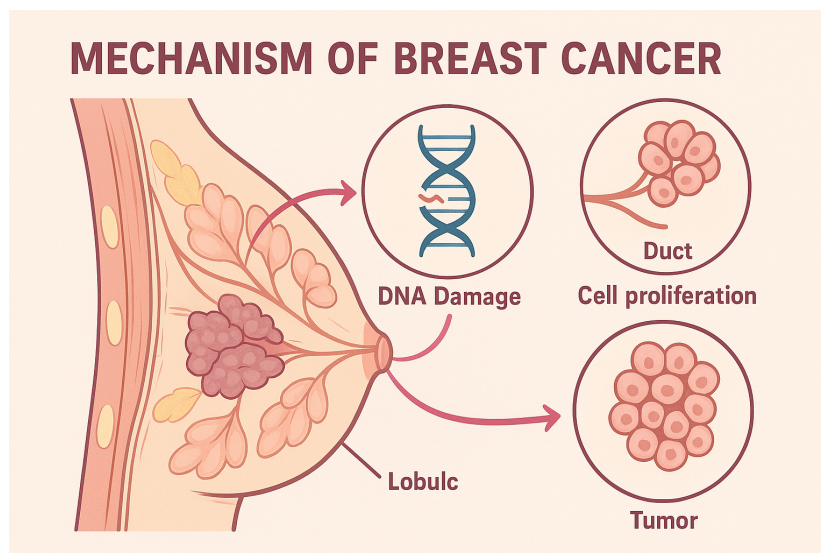


Molecular Testing in Breast Cancer: Clinical Applications and Emerging Technologies

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Abstract

Molecular testing has transformed the management of breast cancer by identifying biomarkers that inform prognosis, predict therapeutic response, and guide personalised treatment strategies. This review delineates the clinical significance of established assays—including immunohistochemistry for ER, PR, and HER2; fluorescence in situ hybridization (FISH) for HER2; and multigene expression profiles such as Oncotype DX, MammaPrint, Prosigna, Endo Predict, and Breast Cancer Index (BCI)—in early-stage, hormone receptor-positive, HER2-negative breast cancer. We discuss actionable genetic alterations, notably PIK3CA mutations, which permit targeted therapy with PI3K inhibitors (e.g., alpelisib) in metastatic settings, as well as BRCA1/2, TP53, and CHEK2 mutations that influence prognosis and treatment decisions. The role of immunotherapy markers—programmed death-



ligand 1 (PD-L1) expression, tumour mutational burden (TMB), and deficiency in mismatch repair (dMMR)—is examined, highlighting their utility in selecting patients for immune checkpoint inhibitors. We also review liquid biopsy

approaches for detecting circulating tumour DNA (ctDNA), which enable early detection of recurrence and dynamic monitoring of treatment resistance. Finally, emerging next-generation sequencing (NGS) panels and novel gene signatures are explored as avenues to further refine personalized care. Through a synthesis of data from at least 15 PubMed-indexed studies, this review underscores the integral role of molecular diagnostics in optimizing clinical outcomes for breast cancer patients.

Background

Breast cancer is the most common malignancy in women worldwide, accounting for approximately 2.3 million new cases and 685,000 deaths in 2020. Advances in molecular biology have revealed that breast cancer is a heterogeneous disease comprised of distinct molecular subtypes—luminal A, luminal B, HER2-enriched, basal-like (often triple-negative), and normal-like—each with unique biological behaviours and clinical outcomes. Traditional clinicopathological features (tumour size, grade, lymph node status) remain important; however, molecular testing provides additional prognostic and predictive value by elucidating tumour biology at genomic and transcriptomic levels.

Immunohistochemistry (IHC) for estrogen receptor (ER), progesterone receptor (PR),

and human epidermal growth factor receptor 2 (HER2)

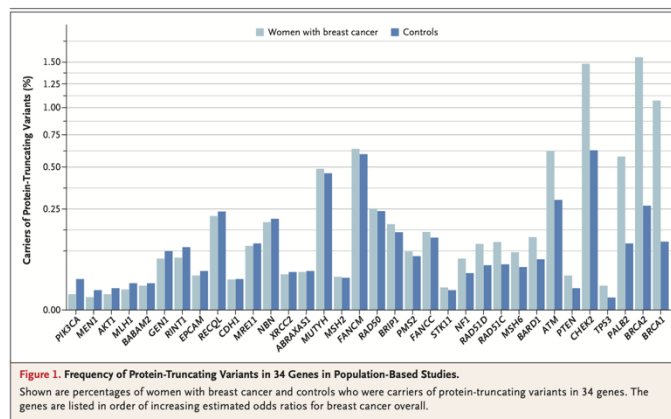
forms the cornerstone of initial molecular evaluation in breast cancer. ER and PR positivity predict response to endocrine therapy, while HER2

overexpression or gene amplification indicates

eligibility for HER2-targeted therapies such as trastuzumab,

pertuzumab, and ado-

trastuzumab emtansine. Accurate determination of HER2 status requires IHC scoring supplemented by FISH when IHC is equivocal (2+), per ASCO/CAP guidelines.

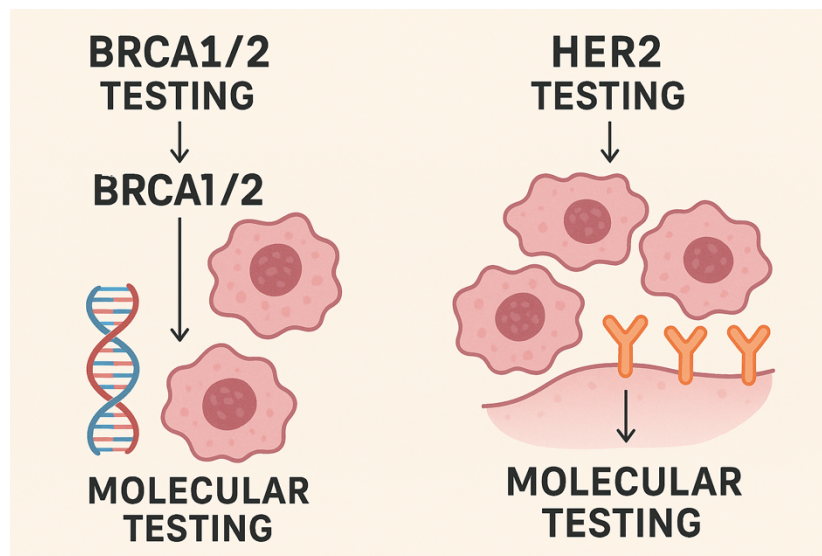


Despite the prognostic value of hormone receptor and HER2 testing, up to 70 % of early-stage, ER-positive, HER2-negative breast cancer patients may not derive substantial benefit from adjuvant chemotherapy, leading to overtreatment and unnecessary toxicity. Multigene expression profiles were developed to refine risk stratification in this subgroup, providing individualized recurrence risk scores and chemotherapy benefit predictions. Furthermore, somatic mutation analysis—particularly for genes such as PIK3CA, BRCA1/2, TP53, and CHEK2—has expanded therapeutic options by identifying candidates for targeted therapies and assessing hereditary cancer risk.

Molecular Markers and Their Clinical Utility

Hormone Receptors and HER2

ER and PR expression, determined by IHC, are the most established predictive biomarkers in breast cancer. ER-positive tumours respond to endocrine therapies such



as tamoxifen, aromatase inhibitors, or selective estrogen receptor degraders (e.g., fulvestrant), which improve disease-free and overall survival compared to placebo or chemotherapy in hormone receptor-positive disease.

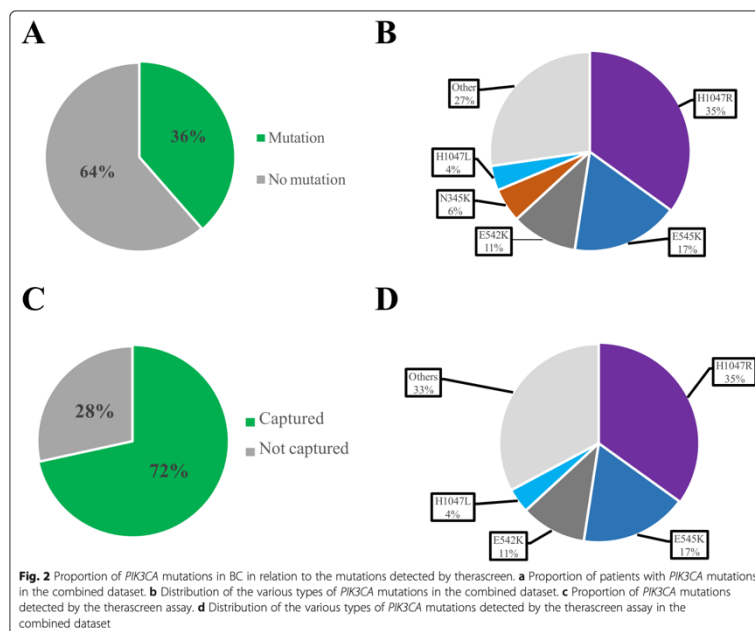
PR positivity, although correlated with ER, provides additional prognostic information and may inform combined endocrine therapy decisions. HER2 overexpression or amplification, identified by IHC (3+) or FISH, is present in approximately 15 %–20 % of breast cancers and confers a more aggressive phenotype. HER2-positive patients benefit from trastuzumab-based regimens, which have substantially improved outcomes in both early and metastatic settings.

Somatic Mutation Analysis

PIK3CA Mutations

PIK3CA encodes the p110 α catalytic subunit of phosphatidylinositol-3-kinase (PI3K), a critical mediator of the PI3K/AKT/mTOR signalling pathway. Somatic PIK3CA

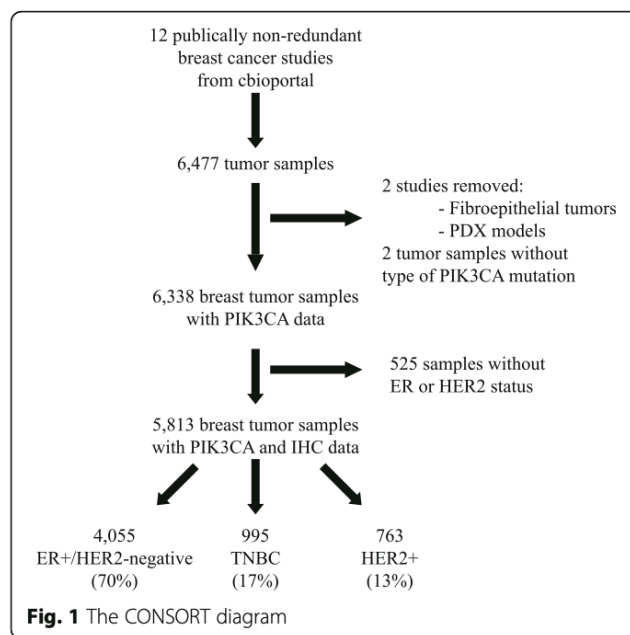
mutations occur in approximately 30 %–40 % of breast cancers, predominantly in luminal A (45 %) and luminal B (29 %) subtypes, and 39 % of HER2-enriched



tumours. In early breast cancer, *PIK3CA* mutations correlate with ER positivity, older age, lower tumour grade, and smaller tumour size. In metastatic hormone receptor-positive, HER2-negative breast cancer, *PIK3CA* mutations are detected in over 36 % of cases and predict sensitivity to PI3K inhibitors such as alpelisib. The phase III SOLAR-1 trial demonstrated that alpelisib combined with fulvestrant significantly prolongs progression-free survival in *PIK3CA*-mutated, hormone receptor-positive, HER2-negative advanced breast cancer compared to placebo plus fulvestrant, leading to FDA approval of alpelisib with the companion diagnostic thetherascreen *PIK3CA* RGQ PCR Kit to detect *PIK3CA* mutations in tissue and/or liquid biopsies.

BRCA1/2 Mutations

Germline BRCA1 and BRCA2 mutations occur in approximately 5 %–10 % of breast



cancer cases and confer high lifetime risks for breast and ovarian cancer. Identification of pathogenic BRCA1/2 variants informs risk-reducing strategies (e.g., prophylactic mastectomy and oophorectomy) and eligibility for PARP inhibitors in metastatic and adjuvant settings. The OlympiAD and EMBRACE trials demonstrated that PARP inhibitors (olaparib, talazoparib) improve progression-free survival in metastatic, HER2-negative breast cancer patients with germline BRCA mutations compared to

chemotherapy. BRCA mutation status also influences surgical decision-making, leading some carriers to choose bilateral mastectomy to reduce recurrence risk.

TP53 and CHEK2 Mutations

TP53 mutations occur in approximately 30 % of all breast tumours and up to 80 % of triple-negative breast cancers, associated with aggressive behaviour and poorer prognosis. While TP53 status correlates with response to certain chemotherapeutic agents and PARP inhibitors, its routine clinical use as a predictive biomarker remains investigational. CHEK2 mutations, such as c.1100delC and I157T, confer a moderate increase in breast cancer risk (approximately 20 % lifetime risk without affected relatives, rising to 44 % with a strong family history). CHEK2 mutations may predict resistance to anthracycline-based chemotherapy and sensitivity to epirubicin; however, their incorporation into clinical decision-making is still under evaluation.

Multigene Expression Assays

Oncotype DX

Oncotype DX is a 21-gene RT-PCR-based assay that yields a recurrence score (RS) ranging from 0 to 100, stratifying early-stage, ER-positive, HER2-negative, and node-

MOLECULAR TESTING IN BREAST CANCER



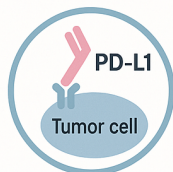
Estrogen Receptor



HER2 gene amplification



Cell proliferation



Immune evasion

negative breast cancer patients into low, intermediate, and high risk for distant recurrence. In the TAILORx trial, 70 % of women with intermediate RS (11–25) showed no additional benefit from chemotherapy when treated with endocrine therapy alone, sparing many women from unnecessary chemotherapy-related toxicity. Retrospective analyses from NSABP B14/B20, SWOG-8814, and other studies validated Oncotype DX's predictive and prognostic utility, demonstrating that patients with

RS < 11 have excellent outcomes with endocrine therapy alone, while those with RS > 26 derive significant benefit from adjuvant chemotherapy. The adoption of Oncotype DX in clinical practice has increased from 1.5 % in 2004 to 34 % in 2015 among ER-positive patients, coinciding with decreased chemotherapy usage among test users and improved survival outcomes compared to non-users.

MammaPrint

MammaPrint is a 70-gene microarray-based assay that categorizes early-stage breast cancer patients as having low or high genomic risk of recurrence, regardless of ER/HER2 status. The MINDACT trial, a prospective, phase III study, demonstrated that 46 % of patients with high clinical risk but low genomic risk (as per MammaPrint) could safely forgo adjuvant chemotherapy without compromising five-year distant metastasis-free survival. MammaPrint also achieved level 1A evidence for clinical validity and utility in the WSG-PlanB, ISPY2, and other trials.

Prosigna (PAM50)

Prosigna, also known as the PAM50 assay, uses a 50-gene signature to classify tumours into intrinsic subtypes (luminal A/B, HER2-enriched, basal) and produces a risk of recurrence (ROR) score for postmenopausal women with lymph node-negative or 1–3 node-positive, hormone receptor-positive breast cancer. The Prosigna ROR score predicts five- and ten-year distant recurrence risk and informs decisions regarding adjuvant chemotherapy and extended endocrine therapy.

EndoPredict and Breast Cancer Index

EndoPredict is a 12-gene RT-PCR assay combined with nodal status and tumour size that yields an EPclin score, stratifying patients with ER-positive, HER2-negative, node-negative or node-positive (1–3 nodes) breast cancer into low- or high-risk categories for recurrence. Prospective validation in GEICAM 9906, ABCSG-6, and ABCSG-8 trials confirmed its independent prognostic value. The Breast Cancer Index (BCI) combines the HOXB13:IL17BR ratio with a molecular grade index, predicting the benefit of extended endocrine therapy and the risk for late recurrence in HR-positive, node-negative breast cancer patients.

Immunotherapy Biomarkers

Immune checkpoint inhibitors have shown efficacy in subsets of breast cancer patients, particularly those with triple-negative breast cancer (TNBC). Biomarkers such as PD-L1 expression, tumour mutational burden (TMB), and microsatellite instability (MSI)/dMMR status identify candidates for immunotherapy. PD-L1 IHC testing uses combined positive score (CPS) methods with antibodies like SP142 and 22C3; the KEYNOTE-355 trial demonstrated that PD-L1-positive metastatic TNBC patients (CPS ≥ 10) derive survival benefit from pembrolizumab plus chemotherapy. High TMB (≥ 10 mut/Mb) correlates with improved response rates to pembrolizumab, as evidenced in the TAPUR study, which reported a 37 % objective response rate in high-TMB tumours across cancer types. MSI/dMMR tumours, though rare (< 2 %) in breast cancer, exhibit high mutational loads and may respond to anti-PD-1/PD-L1 therapies; KEYNOTE-158 showed promising responses in MSI-high

solid tumours including breast cancer. NGS panels that concurrently assess PD-L1, TMB, and dMMR enhance immunotherapy biomarker detection.

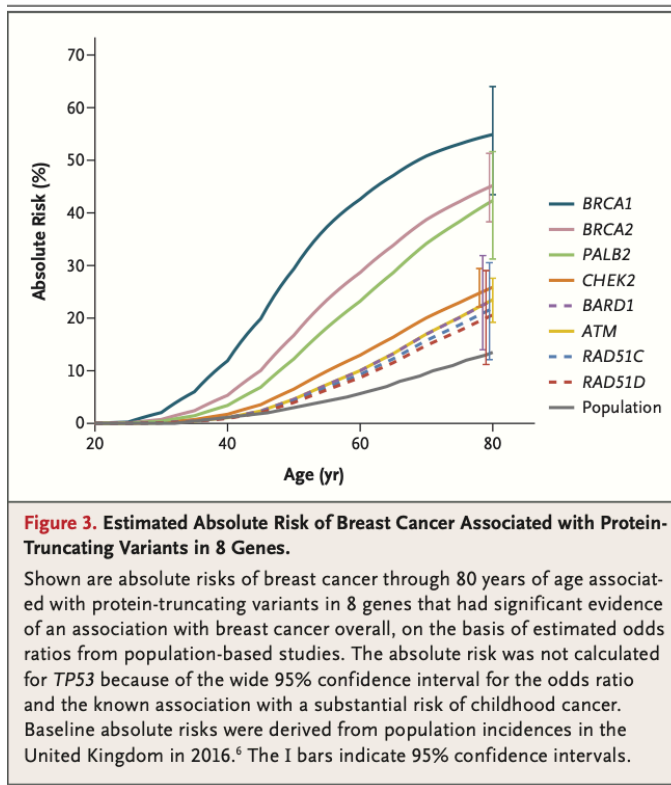
Circulating Tumor DNA (ctDNA) and Liquid Biopsy

Liquid biopsy assays detect ctDNA in plasma, providing a minimally invasive approach to identify actionable mutations, monitor minimal residual disease (MRD), and track the emergence of resistance. The NHS England rollout of ctDNA testing for lung and advanced breast cancer patients represents a global first, enabling the detection of tumour-specific mutations days to weeks before radiographic progression and shortening time to targeted therapy selection by an average of 16 days. The FDA-approved therascreen PIK3CA RGQ PCR Kit detects PIK3CA mutations in ctDNA, guiding alpelisib therapy in PIK3CA-mutated advanced breast cancer. Real-time ctDNA monitoring identifies early relapse and informs treatment adjustments in both early-stage and metastatic settings, with evidence indicating that detectable ctDNA post-surgery predicts impending clinical recurrence by months.

Clinical Impact of Molecular Testing

Prognostic Value

Multigene assays provide prognostic stratification beyond traditional clinicopathological factors. Oncotype DX, validated in NSABP B14/B20, SWOG-8814, and TransATAC trials, predicts five-year distant recurrence risk in early ER-positive, HER2-negative, node-negative breast cancer. Patients with low RS (< 11) have an excellent prognosis with endocrine therapy alone (five-year distant recurrence $< 5\%$), whereas high RS (> 26) indicates high recurrence risk and significant chemotherapy benefit. Oncotype DX use has increased over time, correlating with decreased chemotherapy usage and improved survival outcomes; in the SEER cohort from 2004 to 2015, Oncotype DX utilization rose from 1.5 % to 34 %, associated with an 80 % lower risk of breast cancer-specific death after adjustment for stage and age.



MammaPrint stratifies patients into low- and high-genomic risk groups, independent of ER status; the MINDACT trial revealed that patients with discordant clinical high-risk but genomic low-risk (MammaPrint) can forgo chemotherapy without compromising five-year distant metastasis-free survival (> 94 %). The Prosigna ROR score predicts ten-year distant recurrence risk in postmenopausal, node-negative or 1–3 node-positive, hormone receptor-positive breast cancer, guiding extended endocrine therapy decisions. EndoPredict and BCI similarly stratify

patients for late recurrence risk and extended endocrine therapy benefit.

Predictive Value and Therapeutic Guidance

Endocrine Therapy

ER and PR testing remains the standard for selecting endocrine therapy; ESR1 mutations detected via NGS or ctDNA in metastatic settings predict resistance to aromatase inhibitors, guiding a switch to selective estrogen receptor degraders. Additionally, Oncotype DX RS influences endocrine therapy choice: patients with low RS may require only endocrine therapy, while those with intermediate RS (11–25) may omit chemotherapy if over 50 years old, as in TAILORx.

Chemotherapy

Oncotype DX and MammaPrint directly inform chemotherapy decisions in early ER-positive, HER2-negative breast cancer. In TAILORx, women aged > 50 with RS ≤ 25 had no chemotherapy benefit; women ≤ 50 with RS 16–25 had a small chemotherapy benefit, informing nuanced age- and RS-based recommendations. Meta-analyses estimate that multigene assays reduce chemotherapy use by 30 %–50 % among eligible patients without compromising outcomes.

Targeted Therapy

Testing for PIK3CA mutations via tissue or liquid biopsy identifies candidates for alpelisib in combination with fulvestrant after progression on endocrine therapy. BRCA1/2 mutation testing supports PARP inhibitor therapy (olaparib, talazoparib) in metastatic and high-risk early-stage settings, improving progression-free and overall survival. NTRK fusion testing, although rare (< 1 % in breast cancer), indicates eligibility for TRK inhibitors (larotrectinib, entrectinib), with high response rates observed in NTRK fusion-positive solid tumours including secretory breast carcinoma. PI3K pathway inhibition, and CDK4/6 inhibitors (palbociclib, ribociclib, abemaciclib) benefit ER-positive, HER2-negative metastatic breast cancer and are now moving into adjuvant trials, with biomarker-driven selection under investigation.

Immunotherapy

PD-L1 testing using IHC (CPS ≥ 10) identifies metastatic TNBC patients for pembrolizumab plus chemotherapy, improving overall survival compared to chemotherapy alone. High TMB and MSI/dMMR status, though uncommon in breast cancer, may predict immunotherapy response; NGS panels enable simultaneous assessment, potentially expanding the immunotherapy-eligible population.

Monitoring and Minimal Residual Disease

ctDNA assays detect residual disease and track clonal evolution in real-time. In advanced breast cancer, serial ctDNA monitoring predicts progression months before radiographic changes, allowing early therapeutic intervention. Postoperative ctDNA positivity in early-stage patients correlates with significantly increased relapse risk, suggesting a role for ctDNA in identifying candidates for adjuvant therapy intensification or novel trial enrollment.

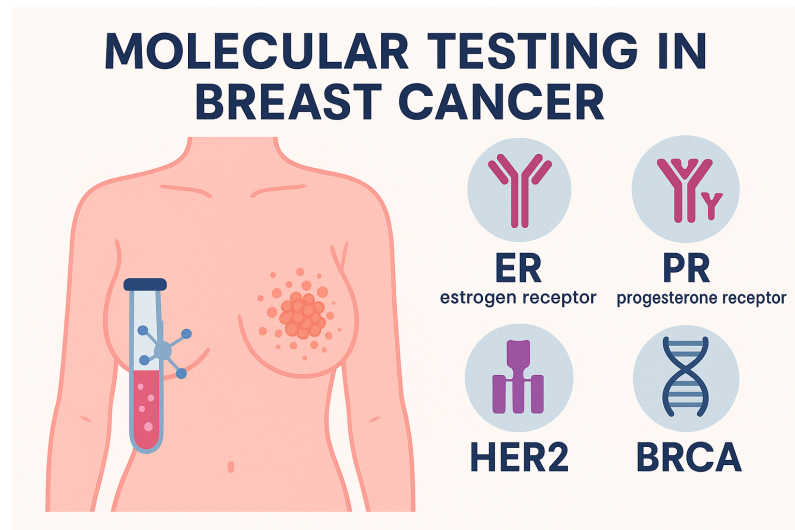
Future Directions

Next-Generation Sequencing (NGS) Panels

Comprehensive NGS panels (e.g., MSK-IMPACT, FoundationOne CDx) profile hundreds of cancer-related genes, identifying actionable mutations, copy number alterations, and gene fusions in a single assay. Their integration into clinical practice enables broad molecular profiling, guiding enrollment in genotype-matched clinical trials and informing off-label targeted therapy use. Challenges include assay standardization, interpretation of variants of unknown significance, and ensuring equitable access.

Emerging Biomarkers and Gene Signatures

Novel gene signatures continue to emerge from transcriptomic and epigenomic research. Assays such as BCI, EndoPredict, and Prosigna refine risk prediction for



late relapse and extended endocrine therapy benefit, addressing clinical gaps left by Oncotype DX and MammaPrint. Circulating tumour cell (CTC) enumeration and analysis, as well as extracellular vesicle profiling, offer complementary liquid

biopsy approaches, though clinical utility is still under evaluation. Integration of artificial intelligence and machine learning on multi-omic data holds promise for discovering novel prognostic and predictive biomarkers.

Personalized Therapy and De-escalation Strategies

Ongoing clinical trials are assessing the de-escalation of therapy in molecularly low-risk patients, including omission of surgery in exceptional responders to neoadjuvant therapy (e.g., NRG-BR005), and tailoring adjuvant regimens based on minimal residual disease detection via ctDNA (e.g., NRG-BR007). Conversely, escalation strategies for high-risk molecular profiles investigate novel combinations (e.g., PI3K inhibitors plus CDK4/6 inhibitors) and exploration of adaptive trial designs to rapidly incorporate emerging biomarkers.

Conclusion

Molecular testing in breast cancer has revolutionized patient management by providing precise prognostic and predictive information that guides personalized therapeutic decisions. Multigene expression assays such as Oncotype DX and MammaPrint refine risk stratification and optimize adjuvant chemotherapy recommendations, reducing overtreatment while maintaining or improving survival outcomes. Somatic mutation analysis for PIK3CA, BRCA1/2, TP53, and other genes informs targeted therapy selection, while immunotherapy biomarkers (PD-L1, TMB, dMMR) expand treatment options in select patient subsets. Liquid biopsy approaches, including ctDNA detection, enable dynamic monitoring of minimal residual disease and early detection of resistance. The integration of comprehensive NGS panels and emerging gene signatures promises further refinement of personalized treatment paradigms. As research continues to unveil novel biomarkers and technologies, molecular diagnostics will play an increasingly pivotal role in achieving precision oncology for breast cancer patients.

Table 2. Risk of Breast Cancer Overall Associated with Rare Missense Variants in 34 Genes in Population-Based Studies and All Studies.

Gene	Population-Based Studies (48,826 patients and 50,703 controls)*			All Studies (60,466 patients and 53,461 controls)*	
	No. of Carriers of Rare Missense Variants		Odds Ratio (95% CI)	P Value	P Value
	Women with Breast Cancer	Controls			
<i>ABRAXAS1</i>	233	242	1.04 (0.86–1.25)	0.70	0.40
<i>AKT1</i>	142	156	0.96 (0.76–1.21)	0.72	0.63
<i>ATM</i>	2411	2471	1.06 (1.00–1.13)	0.051	0.0010
<i>BABAM2</i>	167	170	1.01 (0.81–1.26)	0.91	0.63
<i>BARD1</i>	591	616	1.00 (0.89–1.12)	0.94	0.41
<i>BRCA1</i>	1393	1300	1.11 (1.02–1.20)	0.010	0.027
<i>BRCA2</i>	2831	3038	0.98 (0.93–1.04)	0.50	0.58
<i>BRIP1</i>	868	961	0.95 (0.86–1.04)	0.25	0.54
<i>CDH1</i>	682	668	1.10 (0.98–1.23)	0.096	0.042
<i>CHEK2</i>	895	697	1.42 (1.28–1.58)	2.5×10 ^{−11}	2.9×10 ^{−18}
<i>EPCAM</i>	290	328	0.97 (0.82–1.14)	0.69	0.43
<i>FANCC</i>	597	620	0.95 (0.85–1.07)	0.42	0.80
<i>FANCM</i>	1434	1566	0.95 (0.88–1.02)	0.17	0.85
<i>GEN1</i>	701	707	1.05 (0.94–1.17)	0.38	0.25
<i>MEN1</i>	109	130	0.86 (0.66–1.12)	0.25	0.81
<i>MLH1</i>	677	711	1.02 (0.91–1.13)	0.78	0.68
<i>MRE11</i>	552	611	0.94 (0.84–1.06)	0.33	0.93
<i>MSH2</i>	908	1024	0.92 (0.84–1.01)	0.093	0.12
<i>MSH6</i>	1088	1155	1.00 (0.92–1.09)	0.98	0.74
<i>MUTYH</i>	659	702	1.00 (0.90–1.12)	1.00	0.58
<i>NBN</i>	665	725	0.95 (0.85–1.06)	0.37	0.71
<i>NF1</i>	816	899	0.94 (0.85–1.03)	0.19	0.53
<i>PALB2</i>	805	892	0.96 (0.87–1.06)	0.39	1.00
<i>PIK3CA</i>	170	205	0.83 (0.67–1.02)	0.080	0.33
<i>PMS2</i>	934	963	0.95 (0.87–1.05)	0.31	0.62
<i>PTEN</i>	68	70	1.08 (0.76–1.53)	0.65	0.48
<i>RAD50</i>	1046	1089	0.99 (0.91–1.08)	0.83	0.44
<i>RAD51C</i>	196	206	0.93 (0.76–1.14)	0.49	0.60
<i>RAD51D</i>	224	212	1.05 (0.86–1.27)	0.64	0.57
<i>RECQL</i>	656	627	1.12 (1.00–1.26)	0.047	0.036
<i>RINT1</i>	732	762	1.01 (0.91–1.12)	0.89	0.18
<i>STK11</i>	114	139	0.83 (0.64–1.07)	0.15	0.16
<i>TP53</i>	257	244	1.10 (0.91–1.31)	0.32	0.00080
<i>XRCC2</i>	207	213	1.03 (0.84–1.25)	0.80	0.53

* The sample sizes reflect totals after quality control. The analysis for each gene excluded carriers of protein-truncating variants in that gene.

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